

Simpson's rule

Basic skills

Q1. $a=0$ $b=6$ $h = \frac{b-a}{n}$

a) $h=1$ $n = \frac{b-a}{h} = \frac{6-0}{1} = 6$

b) $n = \frac{6}{3} = 2$ c) $n = \frac{6}{1.5} = 4$

Q2. $h=0.75$ $x_0=3$

a) $x_1 = x_0 + h = 3 + 0.75 = 3.75$

b) $x_2 = 4.5$

c) $x_3 = 5.25$ d) $x_5 = 6.75$

Q3. $f(x) = 2x + 1$ $x_0=1, x_1=3, x_2=5$

a) $y_0 = f(x_0) = 2 \times 1 + 1 = 3$

b) $y_1 = f(x_1) = 2 \times 3 + 1 = 7$

c) $y_2 = 2 \times 5 + 1 = 11$

d) As $x_1 = x_0 + h$ $h = x_1 - x_0$

so $h = 3 - 1 = 2$

e) For $y_5 = f(x_5)$

$x_5 = x_4 + h = 9 + 2 = 11$

so $y_5 = 2 \times 11 + 1 = 23$

f) $\int_1^5 f(x) dx \approx \frac{h}{3} (y_0 + 4y_1 + y_2)$

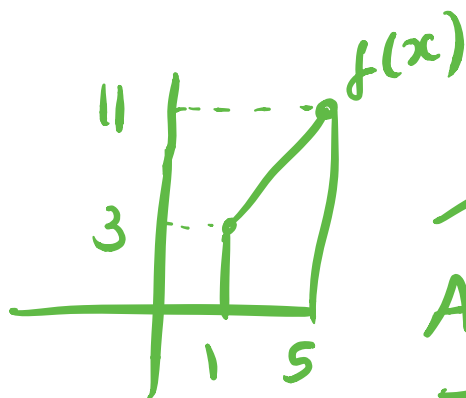
2 strips

$x_0 = 1, x_1 = 3, x_2 = 5$ $h = 2$

$y_0 = 3, y_1 = 7, y_2 = 11$ $n = 2$

so $\int_1^5 f(x) dx \approx \frac{2}{3} (3 + 4 \times 7 + 11)$

$= \frac{2}{3} \times 42 = 28$



Area is $4 \times \frac{1}{2} (3 + 11) = 4 \times 7 = 28$

Exact Simpson's

Q4. $\int_0^2 x \cos^2 x \, dx$ $a=0$ $b=2$
 $n=4$

$$h = \frac{2-0}{4} = 0.5$$

a)

so	$x_0 = 0$	$y_0 = 0 \cos^2 0 = 0$
	$x_1 = 0.5$	$y_1 = 0.5 \cos^2(0.5)$ ≈ 0.385
	$x_2 = 1$	$y_2 = 1 \cos^2 1$ ≈ 0.292
	$x_3 = 1.5$	$y_3 \approx 0.008$
	$x_4 = 2$	$y_4 \approx 0.346$

b) $\int_0^2 x \cos^2 x \approx \frac{h}{3} (y_0 + 4y_1 + 2y_2 + 4y_3 + y_4)$

$$= \frac{0.5}{3} (0 + 4 \times 0.385 + 2 \times 0.292 + 4 \times 0.008 + 0.346)$$

$$= 0.417 \quad (3 \text{ d.p.})$$

c) For exact analytical integration

$$\int_0^2 x \cos^2 x \, dx = \int_0^2 x \left(\frac{1}{2} + \frac{1}{2} \cos 2x \right) dx$$

$$\cos 2x = 2\cos^2 x - 1$$

$$= \int_0^2 \frac{1}{2} x \, dx + \frac{1}{2} \int_0^2 x \cos 2x \, dx$$

↓ by parts $u=x \quad v=\frac{1}{2}\sin 2x$
 $du=1 \quad dv=\cos 2x$

$$= \int_0^2 \frac{1}{2} x \, dx + \frac{1}{2} \left(\left[\frac{1}{2} x \sin 2x \right]_0^2 - \int_0^2 \frac{1}{2} \sin 2x \, dx \right)$$

$$= \left[\frac{1}{4} x^2 \right]_0^2 + \left[\frac{1}{4} x \sin 2x \right]_0^2 + \left[\frac{1}{8} \cos 2x \right]_0^2$$

$$= \left(\frac{1}{4} \times 2^2 - 0 \right) + \left(\frac{1}{4} \times 2 \sin 4 - 0 \right)$$

$$+ \left(\frac{1}{8} \cos 4 - \frac{1}{8} \right) = 0.415 \text{ (3 d.p.)}$$

so Simpson's over estimates

Q5. $\int_0^1 x^3 + 2x^2 + 1 \, dx$

If $h=0.5 \quad n=2 \quad x_0=0, x_1=0.5, x_2=1$
 $y_0=1 \quad y_1=\frac{13}{8} \quad y_2=4$

$$\begin{aligned}
 \text{so } \int_0^1 x^3 + 2x^2 + 1 \, dx &\approx \frac{h}{3} (y_0 + 4y_1 + y_2) \\
 &= \frac{0.5}{3} \left(1 + 4 \times \frac{13}{8} + 4 \right) \\
 &= \frac{23}{12} \approx 1.92
 \end{aligned}$$

If $h = 0.2$ $n = 5$

$x_0 = 0$	$y_0 = 1$
$x_1 = 0.2$	$y_1 = 1.088$
$x_2 = 0.4$	$y_2 = 1.384$
$x_3 = 0.6$	$y_3 = 1.936$
$x_4 = 0.8$	$y_4 = 2.792$
$x_5 = 1$	$y_5 = 4$

$$\begin{aligned}
 \text{so } \int_0^2 x^3 + 2x^2 + 1 \, dx &\approx \frac{h}{3} (y_0 + 4y_1 + 2y_2 + 4y_3 + 2y_4 + y_5) \\
 &= \frac{0.2}{3} (1 + 4 \times 1.088 + \dots + 4) \\
 &= 1.81
 \end{aligned}$$

Exactly

$$\int_0^1 x^3 + 2x^2 + 1 \, dx = \left[\frac{1}{4}x^4 + \frac{2}{3}x^3 + x \right]_0^1$$
$$= \frac{1}{4} + \frac{2}{3} + 1 = \frac{23}{12}$$
$$= 1.92$$

so $h=0.5$ is better (in fact it's exact)

Q 6. $\int_0^1 e^{x^2} \, dx$ $n=5$ $h = \frac{1-0}{5} = 0.2$

$$x_0 = 0 \quad y_0 = e^{0^2} = 1$$

$$x_1 = 0.2 \quad y_1 = e^{0.04}$$

$$x_2 = 0.4 \quad y_2 = e^{0.16}$$

$$x_3 = 0.6 \quad y_3 = e^{0.36}$$

$$x_4 = 0.8 \quad y_4 = e^{0.64}$$

$$x_5 = 1 \quad y_5 = e$$

$$\int_0^1 e^{x^2} \, dx \approx \frac{0.2}{3} \left(1 + 4e^{0.04} + 2e^{0.16} + 4e^{0.36} + 2e^{0.64} + e \right) = 1.317$$

Q7. $h = 0.2$

$a = 0 \quad b = 1$

$n = \frac{1-0}{0.2} = 5$

Have to not use 0.1
as don't have 0.3, 0.5, ..

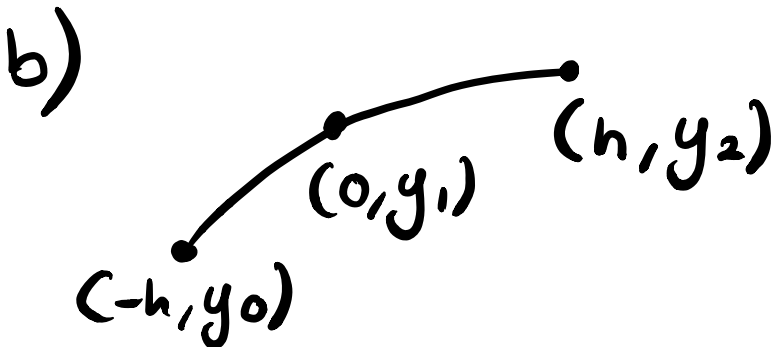
Advanced skills

Q8.

a)
$$\int_{-h}^h Ax^2 + Bx + C \, dx = \left[\frac{A}{3}x^3 + \frac{B}{2}x^2 + Cx \right]_{-h}^h$$

$$= \frac{A}{3}h^3 + \frac{B}{2}h^2 + Ch - \left(-\frac{A}{3}h^3 + \frac{B}{2}h^2 - Ch \right)$$

$$= \frac{2A}{3}h^3 + 2Ch$$



then

Let $y = ax^2 + bx + c$

$y_0 = ah^2 - bh + c$

$y_1 = c$

$y_2 = ah^2 + bh + c$

so $c = y_1$

$$y_2 - y_0 = 2bh$$

$$b = \frac{y_2 - y_0}{2h}$$

$$a = (y_0 + \frac{y_2 - y_0}{2} - y_1) / h^2$$

$$= (\frac{y_0}{2} + \frac{y_2}{2} - y_1) / h^2$$

Hence

$$y = (\frac{y_0}{2} + \frac{y_2}{2} - y_1) \frac{x^2}{h^2} + (\frac{y_2 - y_0}{2}) \frac{x}{h} + y_1$$

c) so Area is

$$\frac{2}{3} \left(\frac{y_0}{2} + \frac{y_2}{2} - y_1 \right) \frac{h^3}{h^2} + 2y_1 h$$

$$= \frac{1}{3} (y_0 + y_2 - 2y_1) h + 2y_1 h$$

$$= \frac{1}{3} (y_0 + y_2 - 2y_1 + 6y_1) h$$

$$= \frac{h}{3} (y_0 + 4y_1 + y_2)$$

← Simpson's rule
with 2 steps
(3 points)

d) By translating

$$\int_{x_2}^{x_4} f(x) dx = \frac{h}{3} (y_2 + 4y_3 + y_4)$$

$$e) \int_{x_i}^{x_{i+2}} f(x) dx = \frac{h}{3} (y_i + 4y_{i+1} + y_{i+2})$$

$$f) \int_a^b f(x) dx = \int_{x_0}^{x_n} f(x) dx$$

$$= \int_{x_0}^{x_2} f(x) dx + \int_{x_2}^{x_4} f(x) dx + \dots$$

Two steps per integration (n is even)

$$+ \int_{x_{n-4}}^{x_{n-2}} f(x) dx + \int_{x_{n-2}}^{x_n} f(x) dx$$

$$\begin{aligned}
 &= \frac{h}{3} (y_0 + 4y_1 + y_2) + \frac{h}{3} (y_2 + 4y_3 + y_4) \\
 &\quad + \dots + \frac{h}{3} (y_{n-4} + 4y_{n-3} + y_{n-2}) \\
 &\quad + \frac{h}{3} (y_{n-2} + 4y_{n-1} + y_n) \\
 &= \frac{h}{3} (y_0 + 4y_1 + 2y_2 + 4y_3 + \dots + 2y_{n-2} \\
 &\quad + 4y_{n-1} + y_n)
 \end{aligned}$$


 Simpson's rule